KRISH KUMAR

AI Engineer | Applied LLM Systems | Full-Stack & Business Automation
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PROFESSIONAL SUMMARY

AI Engineer who ships LLM systems into daily operational use, not demos. I design multi-provider agent architectures — tool calling, RAG over pgvector, multilingual NLU, scheduled autonomous workflows — and build the backend, data and interface layers they depend on. At Upper Crust I own internal engineering for a 12-outlet group and have delivered 40+ production systems in six months, including conversational agents serving ~500 staff over WhatsApp, an MCP server exposing a POS platform as AI-callable tools, document-generation pipelines, and a forensic audit of the group's full transaction history. Earlier, I applied the same stack in fintech: LLM-driven research automation that cut manual analysis time 60%, and ML and statistical pipelines over 5+ years of market data. Strong in Python and TypeScript, and deliberate about correctness in money, auth and data paths.

KEY ACHIEVEMENTS

- Delivered **40+ production systems in six months** as the sole engineer for a 12-outlet restaurant group — POS, ERP, payroll, production planning, e-commerce, CRM and WhatsApp automation.
- Put a **conversational AI assistant into daily use by ~500 staff** over WhatsApp, on an **11-tool agent layer** with per-tool schemas and admin-only gating — the model answers from live operational data, not improvisation.
- Built a **provider-agnostic LLM layer** over Anthropic, OpenAI, Groq and Gemini with a deterministic keyword fallback, so operations survive a degraded provider or an exhausted quota.
- Authored an **MCP (Model Context Protocol) server** exposing a commercial POS platform as schema-validated, AI-callable tools — replacing one-off analysis scripts with a conversational interface over live business data.
- Shipped **six AI features on Groq llama-3.3-70b** in a catering ERP — menu generation, cost estimation, natural-language reports, insights, a design assistant and role-aware chat — each scoped to the user's role, provider key **encrypted at rest**.
- Engineered **autonomous scheduled workflows** — a recurring SOP engine across five frequency modes with primary and secondary escalation owners — so routine follow-up chases itself.
- Took **multilingual NLU into production** across English, Hindi, Hinglish and Gujarati, so frontline staff use the system in the language they actually speak.
- Built **RAG retrieval at 92% relevance** with hybrid search and re-ranking over ChromaDB, plus pgvector knowledge bases answering staff questions from company documents.
- Cut analyst research time **60%** with a multi-provider LLM automation layer, and reduced API spend **35–40%** through intelligent caching and model routing.

TECHNICAL SKILLS

AI / LLM Engineering: Anthropic Claude API, OpenAI, Gemini, Groq, Ollama; multi-provider routing with fallback; prompt engineering, function/tool calling, structured output; RAG with pgvector and ChromaDB; Model Context Protocol (MCP) server authoring; LangChain, LangGraph; Whisper (local ASR); intent classification and multilingual NLU (English, Hindi, Hinglish, Gujarati)

Backend: Python (FastAPI, Django, Flask), Node.js (Express, Fastify), TypeScript; REST API design, webhooks, WebSockets; async SQLAlchemy, Prisma, SQLModel; Alembic migrations; Redis, BullMQ, APScheduler; JWT/NextAuth/bcrypt auth; role- and permission-based access control

Frontend & Mobile: React, Next.js (App Router), Vite, TypeScript, Tailwind CSS, shadcn/ui, Ant Design, TanStack Query, Zustand, Recharts; React Native / Expo; Kotlin with Jetpack Compose, Room, WorkManager, CameraX; Chrome MV3 extensions

Data & Analytics: Pandas, NumPy, SciPy, Parquet, scikit-learn, statsmodels; ETL and reconciliation pipelines; forensic data audit; Playwright and Selenium scraping; ExcelJS, openpyxl, python-docx, python-pptx, PyMuPDF; Matplotlib, Plotly

Databases: PostgreSQL (incl. pgvector), MongoDB, SQLite / better-sqlite3 (WAL), MySQL, Redis

Domain Integrations: Petpooja POS cloud, Tally XML gateway, Razorpay, Shopify, Twilio, Baileys/OpenWA WhatsApp gateways, Google Workspace (Apps Script, Sheets, Drive), Notion, NSE/BSE/MCX market-data and XTS terminal APIs

Platforms & DevOps: Docker and Docker Compose, Git, Linux, AWS, Nginx; Google Apps Script, Sheets, Drive and clasp; Gotenberg; ESC/POS printing; CI with GitHub Actions; Vitest, pytest, Jest, Playwright

ML & Statistical Modelling: scikit-learn, TensorFlow, PyTorch, Hugging Face; feature engineering, time-series analysis, regression and classification, anomaly detection, hybrid recommender systems (collaborative + content-based), computer vision with OpenCV and MediaPipe; walk-forward validation and out-of-sample testing discipline

PROFESSIONAL EXPERIENCE

AI Engineer | Upper Crust (Cafe Uppercrust Group) **May 2026 – Present**
Ahmedabad, Gujarat — 12-outlet bakery, restaurant and catering group

- Own the group's internal engineering: **40+ production systems** across POS, ERP, payroll, production planning, e-commerce, CRM, WhatsApp automation and financial audit — from requirements through deployment and handover documentation.
- Built two **WhatsApp agent platforms**: an employee task bot serving **~500 staff** (FastAPI, PostgreSQL + pgvector, Redis) with @mention task assignment, RAG answers from a company knowledge base, auto-escalation, and NLU across English, Hindi, Hinglish and Gujarati; and an event-driven catering operations agent that schedules SOP steps at signed offsets from an event (T−7d, T+2h) or on recurring windows, collects photo proof and reports on-time performance per roster member.
- Shipped a **109-endpoint multi-session WhatsApp gateway** (Next.js 16, Baileys, Prisma) that exposes WhatsApp as a REST API, so every other internal system sends and receives messages without re-solving pairing, reconnects and session storage.
- Integrated the **Petpooja POS cloud** across the estate — a typed client over 14+ inventory and order endpoints feeding a PostgreSQL store (React + Express + Prisma), plus an **MCP server** that exposes the same APIs as callable tools for AI-assisted analysis.
- Automated the document estate: a **menu automation service** (FastAPI, python-docx, PyMuPDF, Gotenberg) that turns an unstyled Word menu into a designed, photographed client PDF and .docx — with a corpus and photo library that learn from every correction; plus a **human-approved** vendor payment bot that drafts WhatsApp updates from the bank statement and refuses any workbook without a coordinator column.
- Delivered a **forensic margin-leakage audit** across a full year of transaction data using a staged Python/pandas pipeline on Parquet. Quantified recurring leakage, established a high price-integrity rate at the till, and — the finding that mattered — identified material value flowing through controls that did not exist. Shipped a monitoring specification and a runnable control script with it.
- Engineered a **POS-to-Tally accounting bridge** for 12 outlets, replacing a paid vendor connector. Emits balanced Tally XML vouchers per payment mode per day with exact-arithmetic money handling, cost-centre allocation and per-day idempotency, behind 11 test suites.
- Built **k-pos**, an offline-first restaurant POS (TypeScript, Fastify, SQLite WAL, React) replacing the paid SaaS till for core billing — **integer-paise money with basis-point taxes**, price snapshots and transactional order mutations under optimistic concurrency.
- Built a **12-outlet e-commerce platform** (Next.js 14, MongoDB, Razorpay) with outlet-based fulfilment and distance-gated availability, replacing a Shopify subscription.
- Delivered the **HR and payroll stack**: a multi-branch HRMS (React + Express + Prisma monorepo with schemas shared across client and server) plus a zero-cost Google Apps Script estate covering cashbook, ticketing, KYC, payslips and statutory joining forms.
- Architected a **biometric attendance kiosk** (Kotlin, Jetpack Compose, Room, CameraX) where fingerprint templates **never leave the device**: on-device 1:N matching, offline punch queueing, and server-side geofence authority rather than trusting the client.
- Reverse-engineered the incumbent POS web application to specify the replacement, and documented the entire portfolio into a 46-project technical handover with flow diagrams, runbooks and a credential-hygiene remediation.

Research Analyst (AI & Quantitative Systems) | Junomoneta Finsol Private Limited **January 2026 – April 2026**

Applied LLM automation, ML pipelines and real-time data systems for a financial-research desk

- Built the desk's **LLM research automation layer** — Claude, GPT-4, Gemini and Qwen behind a single interface using prompt engineering, function calling and RAG — to summarise research, score news sentiment and generate analyst briefings. **Cut manual research time by 60%.**
- Designed a **multi-model routing and caching strategy** that held output quality while controlling per-query API spend, and drove structured, schema-validated model output so downstream code could consume it without parsing prose.
- Engineered **real-time data pipelines** over WebSocket streams and exchange REST APIs, with automated validation and anomaly detection, feeding a SQLite/Pandas analytical store holding 5+ years of history.
- Developed **statistical and ML models** — time-series analysis, regression, volatility modelling — and a simulation engine that scored each model on out-of-sample performance with **walk-forward validation across 3+ market regimes**, a discipline I still apply to any model claim.
- Built a **real-time analytics application** (PyQt6 desktop + Streamlit web, Plotly) with sub-second streaming, 20+ computed indicators and live risk metrics, **replacing a paid third-party SaaS subscription**; plus 8+ automated monitoring dashboards.
- Applied the modelling work to live systems: algorithmic execution strategies across equity indices and commodities, with documented rules, position sizing and risk limits.

Associate Research Analyst (Data Engineering) | Junomoneta Finsol Private Limited **July 2025 – December 2025**

- Built and operated ETL pipelines processing **100,000+ data points daily** in Python, SQL and Pandas, sustaining **99.9% accuracy** through automated validation and anomaly detection.
- Ran exploratory data analysis across 3+ years of history to surface recurring patterns, clustering behaviour and seasonality that fed downstream model design.
- Prototyped early **AI automation tooling** on Claude and GPT-4 using function calling, custom tools and retrieval, establishing the patterns later productionised across the desk.
- Developed ingestion over WebSocket and REST sources plus Matplotlib/Seaborn/Plotly visualisation layers used daily by the research team.

SELECTED PROJECTS

WhatsApp AI Operations Assistant — *FastAPI, PostgreSQL + pgvector, Redis, APScheduler, Baileys/OpenWA, React*

- Conversational operations agent for **~500 staff** who will not install another app. Admins assign work by @mention in a WhatsApp message; staff reply in natural language to complete it, ask questions, or attach photo and document proof.
- **Agentic tool layer** — 11 registered tools (pending tasks, today's schedule, SOP ownership, team status, department backlog, overdue list, staff lookup) with per-tool argument schemas and admin-only gating, so the model answers from live data instead of guessing.
- **Recurring SOP engine**: definitions scheduled daily, weekday-only, weekly by day-of-week bitmask, monthly by date, or hourly across a time window, each spawning tracked executions with a primary and secondary escalation owner.
- **Multi-provider NLU** across Anthropic, OpenAI, Groq and Gemini with a keyword fallback that keeps the bot usable when no API key is configured; handles English, Hindi, Hinglish and Gujarati. RAG knowledge base over uploaded documents, plus conversation memory, web search, audit logging and automated analytics reports.

Shagun Catering CRM / ERP — *React 19, TypeScript, FastAPI, PostgreSQL 16, Docker, Nginx, Groq llama-3.3-70b*

- Full event lifecycle in one system — inquiry capture, menu planning, presentation design, kitchen production, warehouse dispatch and final settlement — across a parallel-track pipeline: New → Follow Up → Menu + Presentation → Confirmed → Event → Completed.
- **7 role-based dashboards** (admin, sales head, menu planner, presentation exec, operations manager, kitchen, warehouse), each with its own KPIs and AI tools.
- **AI features built on Groq llama-3.3-70b**: menu generator, event cost estimation, natural-language business reports, data-driven insights, a design assistant, and a role-aware chat assistant scoped to each dashboard's context.
- Async FastAPI + SQLAlchemy 2 over asyncpg with Alembic migrations and 25+ domain models; JWT auth with 15-minute access and 7-day httpOnly refresh tokens, and the AI provider key **encrypted at rest**. Deployed via Docker Compose behind Nginx.

Production ERP — Manufacturing Unit — *Next.js, TypeScript, Prisma, PostgreSQL, NextAuth, shadcn/ui*

- End-to-end production control on a **22-model schema**: versioned recipes, production plans exploded into raw-material demand, and plan-versus-actual consumption.
- **FIFO stock layers with an explicit ledger** rather than a mutable quantity column, and dispatch lines allocated back to the specific layer they drew from — which is what makes batch traceability possible.
- Raw-material request fulfilment for outlets, barcode label generation, Excel export, and a role × section × permission matrix with an audit-log row on every mutation.

ERP with Petpooja POS Integration — *React, TypeScript, Express, Prisma, PostgreSQL, Docker*

- Centralised orders and inventory for a 12-outlet group into a database the business owns, replacing per-outlet logins to the POS vendor's own shallow reporting.
- Typed API client over **14+ Petpooja inventory and order endpoints** with pagination, retry and a field-mapping layer; **idempotent upserts** so re-running a sync for the same window never duplicates records.
- Multi-outlet filtering, drill-down detail, and the historical store that downstream analytics and the margin audit read from.

Purchase Comparison & Vendor Award Dashboard — *Next.js, TypeScript, better-sqlite3, ExcelJS, React-PDF*

- Replaced the purchasing team's legacy comparison spreadsheet while **reproducing its totals cell for cell** — a regression test against the source workbook is what made the migration defensible to the buyers.
- Vendor ranking per line item with **per-slab GST**, all derived numbers isolated in a pure, fully tested calculation module with no database access.
- Idempotent ERP master imports matched on vendor name and (description, purchase unit), a data-quality report listing what needs a human decision, and export of a PO-ready order sheet accepted by the Petpooja ordering template.

Sales Analytics Dashboard — *JavaScript, in-browser CSV parsing, Chart.js*

- Single-file dashboard over the group's full order history — no server, no database, no upload, so commercial sales data never leaves the machine.
- Parses a **multi-hundred-thousand-row CSV in the browser** and drives every chart from a five-way filter (outlet, segment, order type, month, payment).
- Derived a bakery-versus-restaurant segmentation from order group names because the export has no line-item column, and documented the limit honestly: mixed orders stay their own bucket and are never double-counted.

Ticket Management System — *Google Apps Script, Google Sheets, React Native / Expo*

- Live support-desk system with an **SLA clock, ON HOLD state and automated escalation**, serving a web app and a React Native mobile client from one backend and design system.
- Hold time is credited **on read**, not only at resume, so a held ticket cannot silently age into breach; ticket IDs are allocated under a lock held through the row append to kill a concurrency race.
- Public-versus-internal comment and attachment separation by role, login rate limiting (5 failures → 15-minute lockout) on a public URL, and an ID-generation fix that cut a full-sheet read down to a single cell read.

Cashbook — Petty Cash & Invoicing — *Google Apps Script, Google Sheets, Google Drive, Android (Kotlin)*

- Multi-business petty-cash ledger with **four-tier role-based access** (owner, accountant, manager, user) and per-user business grants — zero hosting cost, Sheets as the database and Drive for receipts.

- Cash-out entries **require an attached receipt**, enforced server-side; backdating is restricted to owner and accountant; the first-owner bootstrap only runs while zero owners exist, so it cannot be replayed.
- Pure calculation logic separated out for local unit testing, and a Kotlin Android wrapper with a FileProvider so staff capture receipts straight from the phone.

Employee Salary Slip Portal — *Google Apps Script, Google Sheets, Google Drive*

- Self-service portal where employees authenticate with **employee ID and date of birth** — no passwords to issue or reset — and download only their own payslips.
- Columns are resolved **by header name**, so HR can add, remove or reorder sheet columns without breaking the portal; date matching is digits-only so any separator works.
- Monthly operation is dropping correctly named PDFs into a Drive folder: no redeploy, no code change, and the whole workforce's payslips distributed without HR emailing anyone.

Facial Recognition Attendance System — *Python, OpenCV, MediaPipe, scikit-learn, SQLite*

- Live webcam attendance: **MediaPipe BlazeFace detection** with automatic model download, face-crop extraction and an OpenCV recogniser trained on enrolled images.
- Confidence-thresholded matching so an unknown face is rejected rather than misattributed, SQLite-backed enrolment and attendance records, same-day duplicate prevention, and CSV export of the daily register.

EARLIER EXPERIENCE & TRAINING

Python Developer Intern | Prodigy Infotech **2024**

- Built 4 production-ready Python projects and an ETL pipeline for financial data; cut execution time 20% and memory use 15% through refactoring, under Git branching and PEP 8 discipline.

Web Development Trainee | Eshuzo Global Technologies Pvt. Ltd. **September 2021**

- 60-hour intensive in HTML5, CSS3, JavaScript, Django and MySQL; built responsive pages and Django CRUD with authentication.

EDUCATION

B.Tech, Computer Science (Data Science) | Brainware University **2025**

- Coursework: Data Structures, Machine Learning, Artificial Intelligence, Database Management, Statistics, Data Mining, Data Visualization, Big Data Analytics.

Diploma, Computer Science | Govt. Polytechnic Bhaga **2019 – 2022**

- Coursework: C/C++, Data Structures, Database Management, Web Technology, Computer Networks, Operating Systems, Software Engineering.

CERTIFICATIONS

- **NISM Series VIII — Equity & Derivatives** · National Institute of Securities Markets, 2025 — SEBI-mandated certification covering derivatives products, trading mechanics, clearing and settlement, and the regulatory framework governing equity derivatives in India.
- **NSE Academy Certified Financial Markets** · NSE Academy, 2025 — capital-market structure, instrument classes, market operations and risk fundamentals.
- **Bloomberg Market Concepts (BMC)** · Bloomberg, 2024 — terminal-based curriculum across economic indicators, currencies, fixed income and equities.
- **Python Programming for Data Science** · Udemy, 2024 — data wrangling, analysis and visualisation workflows with Pandas and NumPy.
- **The Complete MySQL Bootcamp** · Udemy, 2024 — relational modelling, joins, indexing and query optimisation.
- **The Complete C & C++ Programming Course** · Udemy, 2024 — memory management, pointers, object-oriented design and core data structures.
- **R Programming for Statistical Analysis** · Udemy, 2024 — statistical modelling, hypothesis testing and regression in R.